Neural search ranking model (v2)

Replacing the BM25 baseline ranker with a learned neural model for search results increases search-to-purchase conversion by showing more relevant products on the first page.

AT A GLANCE

Primary metric is *conversion_rate*; decision is binary against Ship if search_to_purchase_rate 95% CI excludes zero AND page_load_time guardrail does not breach (no increase > 5%).

Pre-registered before any analysis. Decision is binary against conversion_rate: a 2.0% relative relative lift over a 10.0% baseline requires 4,800 per arm at the standard a / power settings. Guardrails sit on revenue_per_user.

SAMPLE REQUIRED (PER ARM)

4,800

to detect a 2.0% relative relative effect at a = 0.05, power = 0.80

BASELINE
10.0%

DURATION
14 days

TOTAL SAMPLE
9,600

FEASIBILITY
VIABLE

MECHANISM

Why we think this works

Offline relevance evaluation showed NDCG@10 improvements of 8-12% on held-out search judgments. Online behavior should follow the offline signal.

Search-to-purchase has trended down 2% YoY as the catalog grew. Latency budget for search results is 300ms p95; the neural model adds inference cost.

METRICS

conversion_rate is the verdict-driving metric.

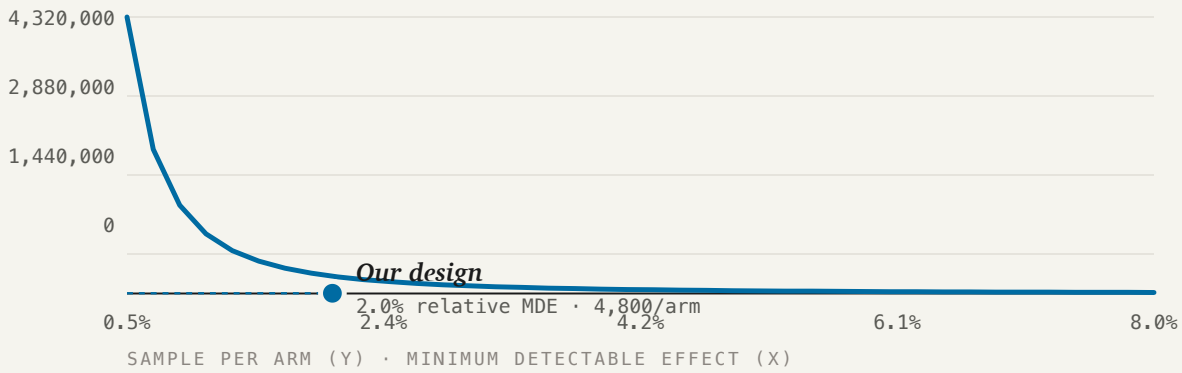
Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.

PRIMARY	**conversion_rate** Proportion · baseline 10.0% · MDE 2.0% relative · direction higher is better
SECONDARY	**add_to_cart_rate** Leading indicator. Not used for the ship/no-ship decision but informs interpretation.
GUARDRAIL	**revenue_per_user** Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative.

POWER

Detecting smaller effects requires much more sample.

To resolve a 2.0% relative relative effect against a 10.0% baseline, we need 4,800 per arm. Tighter detection thresholds would climb the curve to the left — quickly.



Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.

ASSIGNMENT + COHORTS

First-exposure attribution at the user_id level.

Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.

UNIT	**user_id** One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed.
ARMS	**50 / 50** control · treatment

COHORT	all assigned users

DECISION RULES

Pre-registered. The analyze verb walks the same tree against the same metrics.

Ship if search_to_purchase_rate 95% CI excludes zero AND page_load_time guardrail does not breach (no increase > 5%).

SHIP	Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7
LIFT WITH CAVEAT	Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out
NO-SHIP · GUARDRAIL	Any guardrail's 90% CI lies past its harm threshold
NO-LIFT	Well-powered, primary CI straddles zero with half-width > 2 × MDE
INCONCLUSIVE	Underpowered (n < required) AND primary CI straddles zero
INVALID	SRM χ^2 $p < 0.0005$ and no override resolved — investigate assignment